

Summary of info system

Chapter 1

Information technology (IT) can be a powerful tool for a company to gain a competitive advantage. It helps businesses become more focused on their customers, streamline operations, adapt to change, and leverage knowledge.

Building a Customer-Focused Business

The modern economy is driven by improving customer value, and IT is essential for this shift. By using systems like customer relationship management (CRM) and other e-business applications, companies can become more customer-focused. This allows them to keep customers loyal, anticipate their needs, and provide high-quality, personalized service.

For example, Hilton Hotels uses its website, Hilton.com, to provide a fast and personalized reservation experience for guests. The company's goal is to create detailed customer profiles to better understand what they like and don't like, which was not cost-effective before the web. By integrating various systems like workflows, reservation systems, and call centers, Hilton is able to get more specific customer data and provide a single point of contact for travelers.

Reengineering Business Processes

Business process reengineering (BPR) is the radical redesign of business processes to achieve dramatic improvements in things like cost, quality, and speed. This approach can lead to big payoffs but also carries a high risk of failure and disruption.

IT plays a major role in BPR by increasing the efficiency of processes and improving communication and collaboration among employees. Companies often use cross-functional teams and tools like Enterprise Resource Planning (ERP) software to reengineer and integrate their business processes.

* Business Process Improvement vs. Reengineering

* Improvement is an incremental, continuous process that makes small changes to existing processes. It is typically narrow in scope, happens within departments, and requires less time.

* Reengineering is a radical, one-time change that involves starting with a "clean slate." It is broad and cross-functional, takes a long time, and has a high risk.

Johnson Controls Inc. (JCI) successfully used web-based collaboration to reengineer its product design process. By allowing teams to analyze product ideas in the early design stages, the company saved 80% on research and development investments and reduced costs by \$20

million. What once took days of sending packages back and forth now only takes a few hours on the web.

Becoming an Agile Company

An agile company is one that can thrive in rapidly changing markets by producing individualized, high-quality products and services. These companies can offer products tailored to a customer's specific needs while still maintaining high production volumes, a concept known as mass customization.

Agile companies rely heavily on IT to manage their business processes and treat many customers as individuals. They follow four main strategies:

- * They view products as solutions to a customer's problems, which allows them to price based on value, not cost.
- * They cooperate with customers, suppliers, and even competitors to bring products to market quickly and cost-effectively.
- * They use flexible organizational structures to thrive on change and uncertainty.
- * They leverage the knowledge of their employees by nurturing an entrepreneurial spirit.

Dell Inc. is a prime example of an agile, customer-focused company that uses a build-to-order business model. Customers can order a customized computer system online, and it's built only after the order is placed. This process is so efficient that the factory rarely needs more than two hours' worth of parts inventory.

Creating a Virtual Company

A virtual company uses IT to connect people, organizations, and assets. These companies form flexible and adaptable virtual workgroups and alliances to take advantage of fast-changing business opportunities. They use the internet, intranets, and extranets to link with suppliers, customers, and even competitors.

The main strategies behind forming virtual companies include:

- * Sharing infrastructure and risk with partners.
- * Linking complementary skills and knowledge.
- * Reducing the time it takes to get from an idea to a finished product.
- * Gaining access to new markets and increasing market coverage.
- * Migrating from selling products to selling complete solutions.

Building a Knowledge-Creating Company

In a business environment full of uncertainty, the most reliable source of competitive advantage is knowledge. A knowledge-creating company is one that consistently creates new knowledge, shares it widely, and quickly incorporates it into new products and services.

These companies use knowledge management systems (KMS) to help employees share what they know, whether it's explicit knowledge (like documents) or tacit knowledge (the "how-tos" that workers know). The goal of KMS is to help workers create, organize, and access important business knowledge whenever and wherever they need it.

Siemens AG successfully implemented a global knowledge management system called ShareNet. The website acts as a database, chat room, and search engine where employees can share information about successful projects and presentations. This has allowed employees to get help quickly and has resulted in a significant increase in sales, like helping a team in Malaysia win a \$3 million contract by connecting them with a team in Denmark that had relevant expertise.

Chapter 2

An effective way to think about information systems is not just as a tool for daily business operations, but as a strategic asset that can help a company gain a competitive advantage and reinvent itself. A strategic information system is a computer system used at all levels of an organization that helps a company gain an edge over its competitors. These systems fundamentally change a firm's goals, products, services, or how it interacts with its environment.

For example, Citibank's development of ATMs and debit cards in 1977 helped it become the largest bank in the United States. Similarly, State Street Bank and Trust Co. transformed its core business from traditional banking to electronic record-keeping services for securities and mutual funds, helping it to adapt and succeed in a dynamic business environment.

Models for Finding Strategic Opportunities

Two models can help a business identify where to use information systems for competitive advantage: Michael Porter's Competitive Forces Model and the Value Chain Model.

The Competitive Forces Model

This model helps a company understand the five competitive forces it faces:

- * The threat of new competitors entering the market.
- * The bargaining power of customers.

- * The bargaining power of suppliers.
- * The threat of substitute products or services.
- * The rivalry among existing competitors.

To counter these forces, a business can implement five basic strategies using information systems:

- * Cost Leadership Strategy: Produce goods and services at a lower cost than competitors. Strategic information systems can help firms lower their internal costs, allowing them to offer products at a lower price.
- * Differentiation Strategy: Differentiate a firm's products or services from its competitors, or reduce the advantages of its competitors. For example, a company can use an information system to analyze data to improve sales and marketing techniques.
- * Innovation Strategy: Find new ways of doing business, which might involve developing unique products, entering new markets, or making radical changes to business processes.
- * Alliance Strategies: Establish new business partnerships with customers, suppliers, or even competitors. For example, Baxter Healthcare International, Inc. used a "stockless inventory" system to shift all inventory responsibilities to the distributor, which helped it build a strong bond with its customers and gain a competitive edge.
- * Growth Strategies: Expand a company's capacity, enter global markets, or diversify into new products and services using IT.

The Value Chain Model

The value chain model identifies specific business activities where information systems can have the most strategic impact. It views a firm as a chain of basic activities that add value to its products and services.

These activities are split into two categories:

- * Primary Activities: These are directly related to the creation and distribution of a product or service. They include inbound logistics, operations, outbound logistics, sales and marketing, and service.
- * Support Activities: These make primary activities possible. They include organizational infrastructure, human resources, technology, and procurement.

An information system can be used to lower costs or add greater value in these specific activities, giving a firm a competitive advantage. For instance, Wal-Mart uses IT to help suppliers with deliveries to its stores, which lowers the costs of warehousing and inventory.

How Companies Use IT for Strategic Advantage

- * **Building a Customer-Focused Business:** Using Internet technologies, companies can provide high-quality products and services tailored to individual customer preferences. By anticipating their needs and responding to their concerns, companies can build customer loyalty and provide better service.

- * **Reengineering Business Processes:** This involves a fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in cost, quality, speed, and service. Information technology plays a crucial role in making these changes.

- * **Becoming an Agile Company:** An agile company can thrive in rapidly changing markets by producing individualized products in high volumes. They use IT to integrate and manage business processes and treat customers as individuals. Agile companies also cooperate with customers, suppliers, and even competitors.

- * **Creating a Virtual Company:** A virtual company uses information technology to link people, organizations, assets, and ideas. These companies form flexible and adaptable virtual workgroups and alliances to quickly take advantage of new business opportunities.

- * **Building a Knowledge-Creating Company:** This involves consistently creating new business knowledge, spreading it throughout the company, and using it to improve products and services. Companies use knowledge management systems (KMS) to help employees create, organize, and share important business knowledge. A great example is Siemens AG, which used its ShareNet website to help employees share their expertise globally, leading to significant increases in sales.

Chapter 3

The Telecommunications Revolution and the Internetworked Enterprise

The way we communicate and conduct business has been transformed by a revolution in telecommunications, which is the electronic exchange of information over a distance. This revolution has two main components: rapid changes in technology and major shifts in the ownership and control of telecommunication services. A key trend is the merging of computers and communications, which has created something far greater than the sum of its parts.

The concept of the information superhighway describes high-speed digital networks that are national or global in scope and open to the public. This network delivers information, education,

and entertainment, and is having a profound impact similar to that of past innovations like railroads. The most well-known example of this concept is the Internet.

Key Trends in Telecommunications

There are three major trends driving the telecommunications revolution:

- * **Industry Trends:** The telecommunications industry has shifted from a government-regulated monopoly to a competitive, deregulated market. This change, along with the merging of computers and communications, has allowed companies to offer a wide range of services, from phone service and cable TV to Internet access. The explosive growth of the Internet has led to even more new products and services, giving business managers more options for their telecommunications needs.

- * **Technology Trends:** The industry is moving towards open systems with unrestricted connectivity. These systems use common standards for hardware, software, and networking, allowing different computers and devices to easily communicate and share information. A major example is the use of Internet networking technologies like Web browsers, HTML, and TCP/IP. Another significant trend is the switch from traditional analog transmission to digital network technologies, which are much faster, more economical, and have lower error rates. Digital networks can also carry multiple types of information, like data, voice, and video, over the same circuits. Finally, there has been a shift from copper wires to technologies like fiber-optic lines and wireless systems, which offer greater capacity and speed.

- * **Application Trends:** These industry and technology changes have increased the number of possible telecommunications applications. Networks now play a vital role in e-business, e-commerce, and enterprise collaboration, supporting both small and large organizations.

The Internet Revolution and the Networked Enterprise

The Internet has become the largest and most important network of networks, connecting millions of users and systems in over 200 countries. It is constantly expanding and has become a global information superhighway. The Internet has no central computer or governing body; instead, messages are forwarded to their destination using a unique address code. Common standards promoted by groups like the Internet Society ensure that messages can flow freely between different computers and networks.

The Internet has many popular uses, including:

- * Surfing for information, entertainment, or e-commerce.
- * E-mail and instant messaging.
- * Participating in online discussions.

- * Publishing your own work on a website.
- * Buying and selling products through e-commerce.
- * Downloading files like software, music, and videos.

Companies are increasingly using the Internet as a platform for strategic business applications, such as collaboration, customer support, and electronic commerce. The business value of the Internet comes from its ability to:

- * Reduce costs of doing business.
- * Attract new customers and increase loyalty of existing ones.
- * Generate new revenue sources and develop new markets.

Enterprise Networking and its Components

An organization's information architecture includes its hardware, software, telecommunications links, and data files. In enterprise networking, these components are arranged to distribute computing power to individual desktops and to create networks that link the entire company.

The primary model for this is client/server computing, which distributes data and processing power across the enterprise instead of centralizing it.

- * The client is the user's computer, which they interact with directly.
- * The server stores and processes shared data, and handles back-end functions like managing shared databases.

The goal is to make the entire network feel like a single, seamless system to the user.

Intranets and Extranets

* An intranet is a network inside an organization that uses Internet technologies to create an Internet-like environment for information sharing, communication, and business processes. Intranets are protected by security measures and can be accessed by authorized employees. They provide services like:

- * Communications and Collaboration: Improving internal communication with e-mail, discussion forums, and videoconferencing.
- * Web Publishing: Making it easy and cheap to publish and access multimedia business documents like company newsletters and product catalogs.
- * Business Operations: Serving as a platform for critical business applications like inventory control and order processing.

* An extranet is a network link that uses Internet technologies to connect a company's intranet with the intranets of its customers, suppliers, or other business partners. Extranets can be established through secure private network links or the public Internet with encryption. The business value of extranets comes from:

- * Easier and faster access for customers and suppliers to a company's resources.
- * The ability to offer new, interactive services to business partners.
- * Strengthening relationships and improving collaboration with customers and suppliers.

Chapter 4

You asked for a simple summary of the provided content on functional business systems. Here is a breakdown of how information systems are used to support the five main business functions.

The Five Main Business Functions

Businesses use information systems in five major functional areas: marketing, production/operations, human resource management, accounting, and finance.

Marketing Systems

The goal of marketing is to plan, promote, and sell products and services. Information systems help businesses with this by assisting with:

- * Interactive Marketing: This uses the internet to create a two-way conversation between a company and its customers. The goal is to get customers to be active partners in creating and improving products. This is done using tools like websites, web forms, and email.
- * Targeted Marketing: This involves tailoring advertising and promotions to specific groups of people. It uses five components to target an audience:
 - * Community: Ads are customized for people in specific online or geographic communities.
 - * Content: Ads are placed on websites with relevant content, like a car rental ad on a travel website.
 - * Context: Advertising appears only on pages that are relevant to a product or service.
 - * Demographic/Psychographic: Marketing efforts are aimed at specific types of people based on factors like age, income, or marital status.
 - * Online Behavior: Ads and promotions are customized based on an individual's past behavior on a website, often tracked using "cookies".

- * Sales Force Automation (SFA): This uses computers and the internet to help salespeople become more productive. Salespeople can use mobile devices to record data from the field and upload it to the company's network, which helps managers with sales analysis and support.

Production/Operations Systems

This function involves planning and controlling the processes that produce a company's goods or services. All businesses, including transportation, retail, and service companies, use these systems to manage things like inventory and the flow of goods.

- * Computer-Integrated Manufacturing (CIM): This is a concept that uses computers to simplify, automate, and integrate all production and support processes. This helps companies quickly respond to customer needs with high-quality products. Key systems within CIM include:

- * Manufacturing Resource Planning (MRP): This helps plan the materials needed for production and integrates that with production scheduling.

- * Computer-Aided Manufacturing (CAM): These systems automate the production process itself. Examples include:

- * Manufacturing Execution Systems (MES): These monitor and control factory floor operations, tracking things like materials, equipment, and personnel.

- * Process Control: This is the use of computers to control an ongoing physical process, like in a chemical plant or a steel mill.

- * Machine Control: This is using computers to control the actions of individual machines, which is also known as numerical control.

Human Resource Systems

This function handles the recruitment, placement, evaluation, compensation, and development of a company's employees. Information systems support this by helping with personnel needs planning, employee development, and controlling company policies and programs.

- * Traditional Functions: Historically, these systems were used for producing paychecks, maintaining personnel records, and analyzing personnel usage.

- * Modern Functions: Today, human resource information systems (HRIS) also help with recruitment, hiring, performance appraisals, benefits analysis, training, and more.

- * Internet and Intranets: The internet is used for online recruitment and communication with job applicants. Company intranets (internal networks) allow the HR department to provide services around the clock. Intranets also enable employee self-service (ESS), where employees can manage their own information, view benefits, and access training materials.

Accounting Systems

These are the oldest and most widely used business information systems. They are used to record and report business transactions and financial events.

* Operational Accounting Systems: These systems handle daily tasks and historical record-keeping. Examples include:

- * Order Processing: Captures and processes customer orders.
- * Inventory Control: Manages changes in inventory and provides reorder information.
- * Accounts Receivable: Records money owed to the company by customers.
- * Accounts Payable: Records money the company owes to its suppliers.
- * Payroll: Manages employee paychecks and compensation data.
- * General Ledger: Combines data from all other accounting systems to produce financial statements.
- * Online Accounting: Internet technologies are transforming these systems by allowing for online transaction processing with customers and suppliers.

Financial Management Systems

These systems help managers make decisions about financing a business and controlling its financial resources.

- * Key Categories: The major types of financial systems include:
 - * Cash and Investment Management: Manages a company's cash position and short-term securities.
 - * Capital Budgeting: Helps evaluate the profitability of proposed capital expenditures.
 - * Financial Forecasting and Planning: This uses financial models to evaluate a company's performance, determine financing needs, and analyze different financing options. These systems often use "what-if" and "goal-seeking" analysis to explore different scenarios.

Chapter 5

Enterprise business systems are integrated, cross-functional applications that help companies manage their business processes more effectively. Instead of focusing on individual business functions like marketing or manufacturing, these systems support a business's core operations by linking together different departments. This approach allows companies to share

information, improve efficiency, and build stronger relationships with their customers, suppliers, and partners.

Many businesses started by replacing older, function-specific systems with these new integrated applications. The internet, including the World Wide Web, intranets, and extranets, now serves as the technology platform for these systems, helping companies manage the flow of information both internally and with their external partners.

Types of Enterprise Applications

Enterprise applications are designed to support different aspects of a company's operations:

- * Enterprise Resource Planning (ERP): This is the central "backbone" of a business. ERP systems focus on a company's internal processes, such as production, distribution, and finance. They automate and integrate many core functions like sales order processing, inventory management, and human resources.
- * Customer Relationship Management (CRM): CRM systems are all about acquiring and keeping customers through marketing, sales, and service processes.
- * Supply Chain Management (SCM): SCM focuses on building efficient relationships with suppliers to get the products and services a business needs.
- * Partner Relationship Management (PRM): PRM aims to acquire and retain partners who can help a company sell and distribute its products.
- * Knowledge Management (KM): KM applications provide tools that help employees collaborate and make decisions.

The Role of ERP Systems

ERP is a comprehensive system that uses an integrated set of software modules to support a company's internal processes. It provides a real-time, unified view of core business operations like production, order processing, and inventory management. ERP software can track business resources (like cash and raw materials) and commitments (like customer orders) across all departments, regardless of where the data was entered.

Benefits of ERP

- * Increased Efficiency: ERP systems improve internal business processes, leading to better quality and efficiency in customer service, production, and distribution.
- * Cost Reduction: Companies often see lower costs for transaction processing and IT support compared to the older, non-integrated systems they replaced.

- * **Better Decisions:** ERP provides managers with timely, cross-functional information, allowing them to make better decisions more quickly.

- * **Greater Agility:** By breaking down departmental barriers, ERP creates a more flexible organization that can adapt and respond to new opportunities.

The Risks of ERP Implementation

While the benefits are significant, implementing an ERP system is a complex and risky undertaking. Some companies have experienced major failures that led to substantial losses in revenue and profits. For example, orders and shipments were lost, inventory levels were unreliable, and some companies even had to file for bankruptcy.

Common Causes of Failure

- * **Underestimating Complexity:** Companies often underestimate the extensive planning, development, and training required for a new ERP system that will radically change their business processes.

- * **Poor Management:** Failures often occur when companies don't involve employees in the planning stages, try to do too much too quickly, or don't provide sufficient training.

- * **Lack of Testing:** Insufficient data conversion and testing are also major causes of failed projects.

- * **Overreliance on Vendors:** Some companies rely too heavily on the claims of software vendors or consulting firms without doing their own due diligence.

Implementing an ERP system is often compared to a "brain transplant" for a company, highlighting the significant risk of business disruption if not done correctly.

Chapter 6

You've provided a lot of information about enterprise business systems. Let's break down the key concepts in a way that's easy to understand.

Customer Relationship Management (CRM)

What it is: CRM is a system that uses technology to help a business manage all its interactions with customers. The main goals are to give employees a single, complete view of every customer and to give the customer a single, complete view of the company. Think of it as a central hub for all customer-related information.

How it works: CRM systems integrate and automate many customer-facing processes across sales, marketing, and customer service. It captures data from every interaction, like phone calls,

emails, and website visits, and stores it in a common database that everyone in the company can access.

Key functions:

- * **Contact and Account Management:** Tracks data about every past and future interaction with customers.
- * **Sales:** Provides sales reps with tools and data to manage their activities, like product information, sales quotes, and real-time customer account history.
- * **Marketing and Fulfillment:** Helps automate direct marketing campaigns, manage customer responses, and analyze the effectiveness of marketing efforts.
- * **Customer Service and Support:** Gives service reps real-time access to customer data to help them manage service requests and resolve problems.
- * **Retention and Loyalty:** Analyzes customer data to identify, reward, and market to the most profitable and loyal customers.

Why companies use it:

- * It costs six times more to acquire a new customer than to sell to an existing one.
- * Increasing customer retention by just 5% can boost profits by 85%.
- * The odds of selling to an existing customer are 50%, compared to just 15% for a new one.

Potential problems:

Many CRM projects fail because of a lack of preparation and understanding. Companies often rely on the technology to solve problems without first changing their business processes or preparing their employees and customers for the new system. In fact, one report found that over 50% of CRM projects did not deliver the promised results.

Supply Chain Management (SCM)

What it is: SCM is a system that helps a company get the right products to the right place at the right time, in the right quantity, and at an acceptable cost. It manages the network of business relationships with suppliers, customers, and partners to create a fast, efficient, and low-cost flow of products from concept to market.

How it works: SCM uses IT to support and manage the connections between a company's key business processes and those of its suppliers, customers, and partners. Many companies use Internet technologies to streamline these processes and improve coordination.

Why companies use it:

- * Faster, more accurate order processing.
- * Reductions in inventory levels and lower transaction costs.
- * Quicker time to market.
- * The ability to be more agile and responsive to customer demands.

Potential problems:

Developing effective SCM systems is complex and difficult. Common problems include:

- * A lack of proper demand planning tools and knowledge.
- * Inaccurate business data, such as production or inventory information.
- * A lack of collaboration among different departments within a company and with outside partners.
- * The SCM software itself can be immature and difficult to implement.

Enterprise Application Integration (EAI)

What it is: EAI is software that connects major business applications, such as CRM and ERP (Enterprise Resource Planning). It acts as a go-between, or middleware, that allows these different systems to exchange data and work together seamlessly.

Why companies use it:

- * It improves customer service by integrating all the customer and product data that reps need.
- * It streamlines sales order processing, so products and services can be delivered faster.
- * Overall, it provides real business value by helping a company respond quickly and effectively to business events and customer demands.

Enterprise Collaboration Systems (ECS)

What it is: ECS are cross-functional information systems that improve communication, coordination, and collaboration among business teams and workgroups.

How it helps: The goal is to make it easier for people to work together by helping them:

- * Communicate: Share information with each other.
- * Coordinate: Align their individual work efforts and resources.

- * Collaborate: Work together on joint projects.

These systems often include software for electronic communication (like email), electronic conferencing, and collaborative work management.

Chapter 7

The World of E-commerce

E-commerce is a comprehensive process that includes the entire online journey of developing, marketing, selling, delivering, servicing, and paying for products and services. It changes the way companies compete and interact with customers and suppliers. Today, many businesses, both large and small, are involved in some form of e-commerce.

Key E-commerce Processes

E-commerce involves a wide range of business processes, from marketing and buying to selling and providing customer service. These processes are supported by Internet-based technologies and various e-commerce applications.

The key processes of e-commerce include:

- * Marketing and Discovery: This involves activities like interactive marketing and market research to help customers find products.
- * Transaction Processing: This covers the steps from ordering and payment to billing and order receipt.
- * Service and Support: This includes providing customer support and other services after a product has been sold.

Essential E-commerce Systems and Technologies

Successful e-commerce operations rely on a suite of essential processes, which can be thought of as a foundation for a company's e-commerce initiatives.

- * Access Control and Security: This ensures trust and secure access by authenticating users and enforcing security measures. This can be done through user names, passwords, encryption keys, and digital certificates. These processes also protect against threats like hacker attacks and theft of passwords or credit card numbers.
- * Profiling and Personalizing: This involves gathering data about your behavior and preferences on a website to create a personal profile. This data is used to give you a personalized view of the site, including product recommendations and targeted ads.

* Search Management: To help customers find what they're looking for, e-commerce sites use search processes that can be based on product descriptions or specific parameters like price ranges.

* Content and Catalog Management: This involves software that helps companies create, update, and manage the information and multimedia content on their websites, often in the form of product catalogs. This software can also work with profiling tools to personalize what each user sees on the site.

* Workflow Management: This uses software to manage and automate business processes. The software contains models with predefined rules and tasks to ensure that the correct data and documents are routed to the right people.

* Event Notification: E-commerce systems are event-driven, meaning they respond to various actions, like a new customer visiting the site or a product being shipped. Event notification software works with workflow management and profiling software to automatically inform all involved parties of important events.

* Collaboration and Trading: These processes support the necessary collaboration between customers, suppliers, and other stakeholders. This includes tools like email and discussion groups to build online communities, as well as trading services and portals for business buyers and sellers.

* Electronic Payment Processes: This is a critical part of e-commerce due to the complexity and security issues of online transactions. Most e-commerce websites use credit card payments. Secure payment methods, such as the Secure Socket Layer (SSL), encrypt data to protect it from being intercepted. Another method is the digital wallet, where your credit card data is encrypted so that only the bank authorizing the transaction can see it, not the merchant. The Secure Electronic Transaction (SET) standard extends this digital wallet approach, but it has not been widely adopted due to increased costs.

Categories of E-commerce

E-commerce is generally divided into three main categories:

* Business-to-Consumer (B2C): This is when businesses sell products and services directly to consumers through attractive online marketplaces.

* Business-to-Business (B2B): This involves electronic marketplaces and direct links between businesses. Examples include secure e-commerce websites for business customers and suppliers, as well as online auctions and exchange portals.

* Consumer-to-Consumer (C2C): This is when consumers buy and sell with each other, such as on auction websites like eBay.

Chapter 8

E-commerce is a key driver for many industries and has changed how companies do business. There are many ways to approach it, and companies often find themselves at a crossroads when deciding on a strategy. One of the most common types is business-to-consumer (B2C) e-commerce, which aims to attract customers, sell products, and build loyalty through good service and engaging features.

To build a successful B2C business, it's crucial to offer attractive products with great value and have a realistic business plan for profitability. Despite the failures of some early dot-com companies, millions of businesses continue to move at least part of their operations to the web.

E-commerce Success Factors

Since a customer is only a click away from a competitor's site, successful e-commerce businesses must focus on building customer satisfaction and loyalty. Key factors for success include:

- * Selection and Value: Offer a good selection of attractive products at competitive prices, along with satisfaction guarantees and customer support.
- * Performance and Service: The website must be fast and easy to navigate, shop on, and purchase from. Customer service should be friendly, helpful, and efficient.
- * Look and Feel: The site should be visually appealing, offering an attractive storefront and product catalogs that provide a good shopping experience.
- * Advertising and Incentives: Use targeted online ads and emails, as well as incentives like coupons and special offers, to attract customers and encourage them to return.
- * Personal Attention: Use customer data to create personalized experiences, such as personal web pages and product recommendations, to encourage repeat visits and purchases.
- * Community Relationships: Build virtual communities for customers through forums, chat rooms, and other collaboration tools to foster a sense of belonging and loyalty.
- * Security and Reliability: Customers must feel confident that their personal and financial information is secure. The business must be trustworthy, delivering on promises for products and shipping.

Key Website Requirements

Successful e-tailers must develop, operate, and manage their websites to become a top destination for consumers. This involves three main areas:

- * **Building the Store:** This includes using website design tools and templates, and often involves hiring an outside contractor or using a web hosting service to build a custom site.

- * **Serving Customers:** Once the site is built, it must welcome and serve visitors efficiently. This means creating user profiles, providing a dynamically updated product catalog, and using an integrated shopping cart system. It also involves offering customer support through online help, email, and discussion groups.

- * **Managing Operations:** The website needs to be managed as a business. This involves analyzing traffic, sales, and inventory with management reports. The website itself must be available 24/7, with sufficient network capacity and security measures like passwords, encryption, and firewalls to protect customer data.

Business-to-Business (B2B) E-commerce

B2B e-commerce is the wholesale side of the business world, where companies buy, sell, and trade with other businesses. All the success factors for B2C also apply to B2B websites. These systems are often integrated with a company's other systems, such as supply chain management and accounting, to ensure all activities are supported by up-to-date data.

Electronic Data Interchange (EDI) is an early form of e-commerce that automates the electronic exchange of business documents like purchase orders and invoices between a company and its suppliers or customers. It tracks inventory, triggers orders, and schedules delivery and payment, which helps streamline processes, save time, and increase accuracy.

Clicks and Bricks Strategies

This is a strategy that bridges a company's physical and virtual worlds. Companies must decide how closely to integrate their e-commerce initiatives with their traditional operations. The main benefit of integrating is leveraging a company's established brand, sharing information, and gaining distribution efficiencies and joint buying power.

There's a spectrum of choices, from full integration to complete separation.

- * **In-house Division (Integration):** Companies like Office Depot fully integrate their e-commerce with their physical stores and catalogs. This allows customers to shop online and pick up purchases in-store, and it helps increase traffic to their physical locations.

- * **Joint Venture:** A joint venture like Kbtoys.com is an e-commerce company created by two separate businesses. While it has independent management, it benefits from shared branding and allows customers to return online purchases to physical stores.

* Strategic Partnership: A less integrated approach, such as the partnership between Rite-Aid and Drugstore.com, where one company owns a smaller stake in the other. They share benefits like joint buying power and integrated distribution but operate with separate management and brands.

* Spin-Off (Separation): A company like Barnesandnoble.com was created as a completely separate e-commerce company. This provides flexibility and access to venture funding but can result in the loss of strategic business opportunities if not integrated with the physical stores.

There is no one-size-fits-all strategy for every company. The decision depends on whether a company's traditional business can provide the resources and capabilities to support a profitable e-commerce model.

Chapter 9

E-commerce websites, such as those used by B2C (business-to-consumer) retailers, must be designed to attract repeat customers. These sites need to be easy to use and manage to be successful.

The key requirements for a successful web store can be broken down into three main categories: developing the web store, serving customers, and managing the store.

Developing Your Web Store

To create a web store, you first have to build an e-commerce website. Many businesses use simple design tools and pre-made templates provided by their web hosting service to build their site, while larger companies may hire software developers or a contractor to create a custom site. Once the website is built, it must be marketed to attract visitors and turn them into loyal customers. Marketing can include:

- * Web page and email promotions
- * Advertising exchanges with other websites
- * Registering the website with major search engines to make it easier for people to find
- * Affiliating with large web portals or e-tailers, such as Yahoo! or Amazon

Serving Your Customers

Once your web store is up and running, it needs to be able to serve customers personally and efficiently to make them loyal buyers. This involves:

- * Creating a personal experience: Use tools to create user profiles and personal web pages or promotions to build a one-to-one relationship with customers. This can be done by encouraging

visitors to register, using "web cookie files" to identify returning visitors, or analyzing a shopper's behavior and preferences on the site.

- * Having a user-friendly design: Your website should have a fast search engine, a convenient shopping cart system, and an updated product catalog with multimedia. The order processing software should be fast and flexible enough to handle personalized promotions, various payment options, and different shipping and tax calculations.

- * Providing support: Offer features like online help menus, tutorials, and frequently asked questions (FAQs) for self-help. You can also provide more personal assistance through email correspondence with customer service representatives, or by setting up discussion groups and chat rooms for customers and staff to interact.

Managing the Web Store

Managing a web store means treating it as both a business and a website. Web hosting companies often provide software and services to help with this, including:

- * Business management: Access to reports that track website traffic, sales, and inventory. Some e-commerce software can also link to accounting programs to help with bookkeeping.

- * Website operation: Ensuring the website is available 24/7. This requires having enough network capacity for peak traffic and backup servers in case of failures. Many hosting services also offer around-the-clock technical support to help with any issues that come up.

- * Security: Using passwords and encryption to protect customer records and transactions. This also includes using firewalls and security monitors to prevent hacking and other threats.